

Lentiviral transduction of iPSCs with sgRNAs and sgRNA libraries

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Reference:

Tian et al (2019). CRISPR Interference-Based Platform for Multimodal Genetic Screens in Human iPSC-Derived Neurons. Neuron pii: S0896-6273(19)30640-3. [Epub ahead of print] PubMed PMID: 31422865.

Reagents

Component	Cat. No	Manufacturer
DMEM (Complete)	11965092	Gibco (Life Tech/Fisher)
TransIT Lenti Reagent	MIR 6600	Mirus Bio LLC
Opti-Mem Reduced Serum	31985070	Gibco (Life Tech/Fisher)
Lentivirus Precipitation Solution	VC125	Alstem LLC
12mL Syringe	4100-X00V0	AIR-TITE (Norm-Ject)
0.45um Filter	SLHV033RB	Millipore Sigma (old: EMD Millipore)

Day 0: Seeding

1. 18-24 hours before transfection, seed 293T cells into a 6 well plate or other format with a density that will make the cells 80-95% confluent on the day of transfection. Refer to a seeding chart if necessary to seed appropriate density.
2. Incubate overnight.

Day 1: Transfection

1. Change 293T media with fresh DMEM.
2. Warm TransIT-Lenti Reagent to RT and vortex gently before using.
3. Gather Opti-Mem, DNA, and packaging mix and refer the table below for recommended amount of reagents to add based on the format of 293Ts that you seeded. Amounts refer to each well of a plate.
4. Typically for individual sgRNAs, I will use 2 wells of a 6 well plate per sgRNA and for sgRNA libraries, I will use 15cm dish per library for a library up to 50,000 elements (can scale down for smaller libraries). Volume for 15cm dish is not listed in table but it is 3x the amount of the 10cm dish.

Culture vessel	48-well plate	24-well plate	12-well plate	6-well plate	10-cm dish	T75 flask
Surface area	1.0 cm ²	1.9 cm ²	3.8 cm ²	9.6 cm ²	59 cm ²	75 cm ²
Complete growth medium	263 µl	0.5 ml	1.0 ml	2.0 ml	10 ml	15 ml
Serum-free medium	26 µl	50 µl	100 µl	200 µl	1.0 ml	1.5 ml
Transfer DNA (1 µg/µl stock)	0.13 µl	0.25 µl	0.5 µl	1.0 µl	5 µl	7.5 µl
Packaging DNA Premix (1 µg/µl stock)	0.13 µl	0.25 µl	0.5 µl	1.0 µl	5 µl	7.5 µl
<i>TransIT</i> [®] -Lenti Reagent	0.78 µl	1.5 µl	3 µl	6 µl	30 µl	45 µl

5. Add Opti-Mem into a sterile tube.
6. In another tube, mix your packaging mix and your DNA.
7. Add your DNA mix to the Opti-Mem and mix gently.
8. Add *TransIT*-Lenti Reagent to the mixture and mix gently.
9. Incubate for 10 min for transfection complexes to form.
10. Add all of the complex mixture to your 293Ts dropwise and gently swirl to mix.
11. Incubate for 2 days. You can check if your cells are making virus by checking fluorescent after 24 hours.

Day 3: Harvest

1. With a 12mL syringe, take up the media/supernatant of your cells.
2. Put a 0.45 µm filter on the syringe and filter the supernatant into a fresh 15mL conical tube. Change filter if it becomes hard to push. Do not push too hard that bubbles are coming out.
3. Add 1:4 ratio of viral precipitation solution. E.g. 0.25mL for 1mL. Mix well by pipetting up and down 10x.
4. Incubate the viral supernatant at 4C for at least 4 hours and up to 3 days but no more than 3 days.
5. Cool down the centrifuge to 4C.
6. Spin down your viral supernatant for 30 min at 1500g.
7. The pellet will contain your virus. Resuspend the pellet with 1mL of your media of choice.
8. Virus can be aliquoted and stored at -80C for long term or 4C for short term (a few days). Flash freeze in liquid nitrogen may increase the retention of their potency.

Transduction with virus

1. Reseed your cells so that they will reach 50% confluent the next day.
2. Add virus to your cells. The amount that you add depends on how concentrated your virus is. I usually add ¼ or ½ of the total virus to the cells.
 - a. For a library prepared from at 15 cm dish of virus, typically infect 10 million iPSCs with 50% of the virus in a matrigel-coated T175 flasks.

- b. For single sgRNAs prepared from 2 wells of a 6-well plate, infect 0.5 million iPSCs in one well of a 6-well plate with 25% of the virus
- 3. Check next day for fluorescence by microscopy, and the next time they are passaged by flow cytometry, to check transduction efficiency.
- 4. For sgRNA constructs including puromycin resistance, add 0.8ug/ml puromycin to select for cells with the sgRNA until they are at least 80% confluent (typically within 2 passages).